



CAEfatigue Robust Design

Product overview

Uncertainty exists in everything from dimensions to material properties, loads, and boundary conditions. The incorporation of these uncertainties leads to models of unprecedented levels of realism and credibility.

CAEfatigue Robust Design enables you to efficiently perform stochastic simulations on both Solver and Fatigue parameters. This allows you to credibly evaluate a broad range of operational scenarios that your product may face in reality. In effect, CAEfatigue Robust Design helps simulate reality and can also work with other FEA solvers when evaluating only durability/fatigue life performance through a CAEfatigue analysis.

Stochastic simulation with a finite element model starts by specifying tolerances on input variables used in the model along with tolerances for variables used in the stress, modal or fatigue analysis. Each tolerance is defined by certain engineering limits and a particular distribution function. The model is then run many times (100 is typical), randomly changing each of the input variables within its assigned tolerance. The result is a cloud of points, referred to as a meta-model, which carries enormous amounts of information and insight.

Features

Model uncertainty in parameter values and system environment.

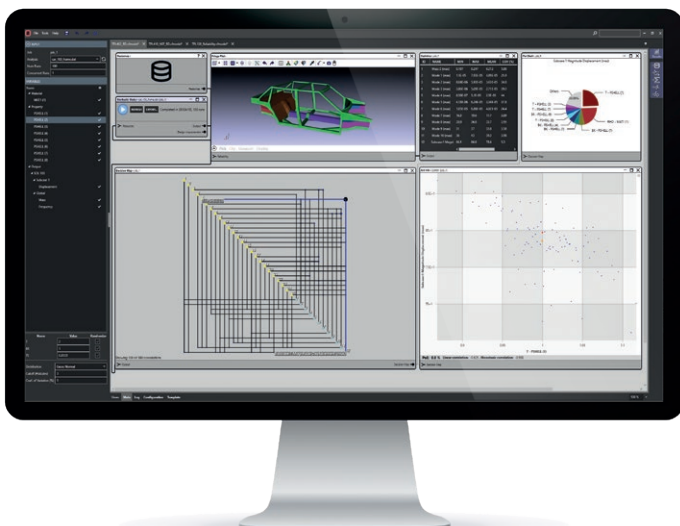
Rapid assignment of tolerance information to thousands of input variables through model information tree.

Combines the power of advanced Monte-Carlo techniques with an **easy-to-use interface**.

Efficient design-to-target via **Stochastic Design Improvement**. Stochastic Design Improvement enables you to design systems to user-specified performance. The cost of this operation, including tolerances, is independent of the number of design variables.

Tight integration with MSC Nastran

- SOL 101 - Linear Statics
- SOL 103 - Eigenvalue Analysis
- SOL 105 - Linear Buckling
- SOL 108/111 - Frequency Response
- SOL109/112 – Transient Analysis
- SOL 600 - Implicit Nonlinear Analysis
- Combined SOL 101, 103, 105 and 108/111
- Including (but not requiring) fatigue response



Case studies

Leverage investment in finite element modeling

Realistic finite element models represent a valuable asset in a company which is an investment that captures corporate know-how and experience. Stochastic simulation provides an opportunity to not only consolidate this knowledge, but also to extract additional, often unexpected, information about a system and its performance against stress and durability results. Meta-models enable engineers to discover and learn about their products in ways that would otherwise require years of experience and testing.

Understand system behavior

CAEfatigue Robust Design offers several ways to navigate meta-models. Shown in the above image is an example of an interactive Decision Map which portrays correlations between input and output variables and the flow of information within a system. A decision map gives structure to information, and structured information is knowledge.

Typical use cases

CAEfatigue Robust Design can easily **detect unusual behaviors** such as the existence of outliers or solutions clusters.

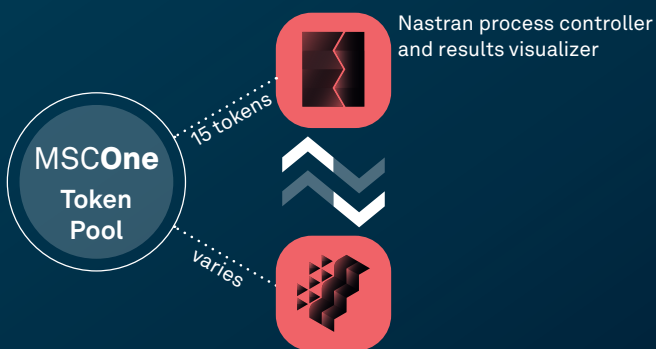
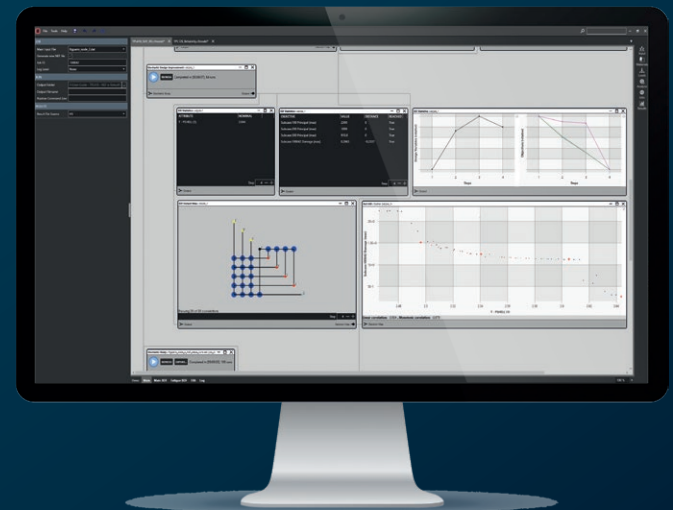
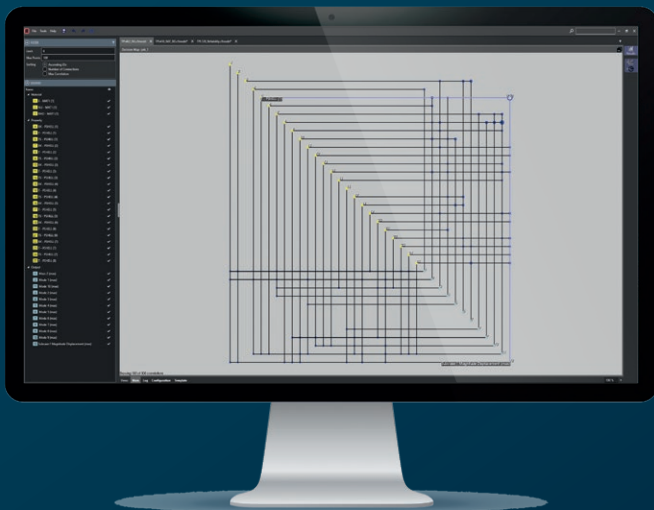
The more complex a product is, the higher the chance that some combination of uncertainties will have an unexpected and potentially undesired effect. This phenomenon, which can only be observed if the model and durability assessment incorporates tolerances that generate results which in many cases can be translated directly into risk and liability. **Understand and manage risk** by evaluating the effect of parameter uncertainty on design performance – this is the key to better designs.

Reduce warranty and product liability costs by identifying and correcting for potentially dangerous outliers using stochastic simulation techniques.

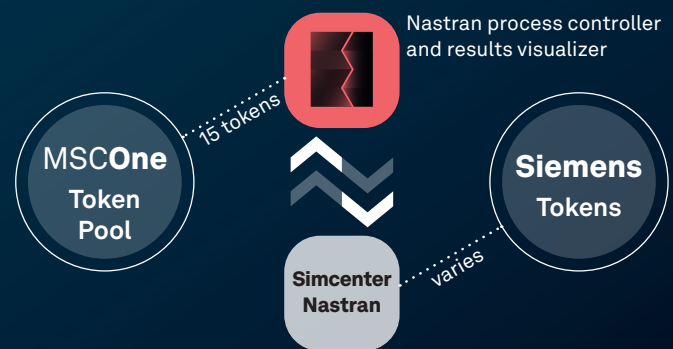
Improve design quality and control costs by understanding which variables drive system behavior.

Improve product performance in the presence of uncertainty with stochastic design improvement techniques.

Gain confidence in your CAE results with simulations that closely mimic the uncertainties seen in reality.



CAEfatigue with MSC Nastran



CAEfatigue with Simcenter Nastran